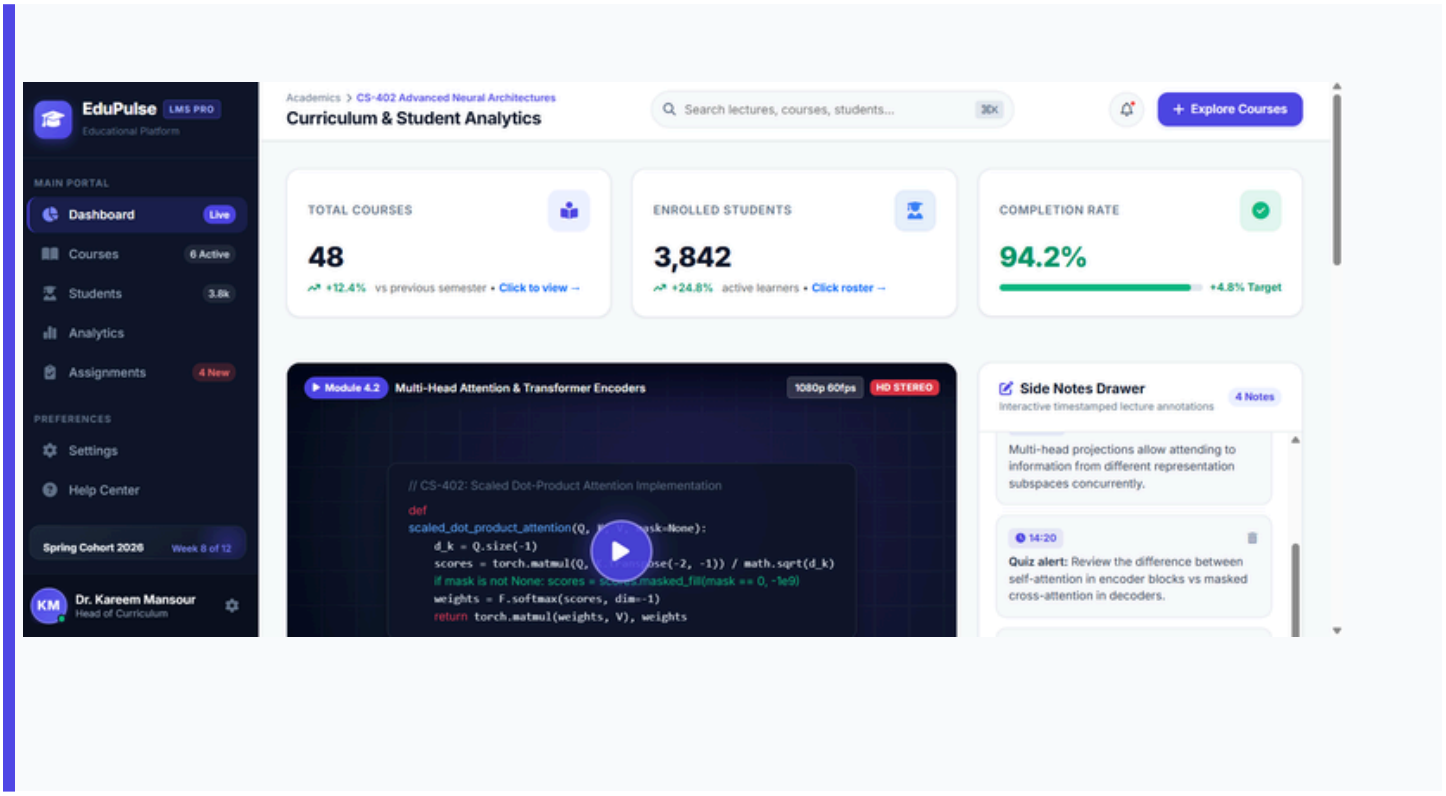
4. UI/UX Design & Prototyping

4.1 Wireframes & High-Fidelity Mockups

The user interface architecture for EduPulse LMS PRO is structured to deliver an intuitive, low-friction learning and administrative ecosystem. High-fidelity visual mockups were crafted to validate screen flow, spatial layout, and responsive UI components across key operational views:

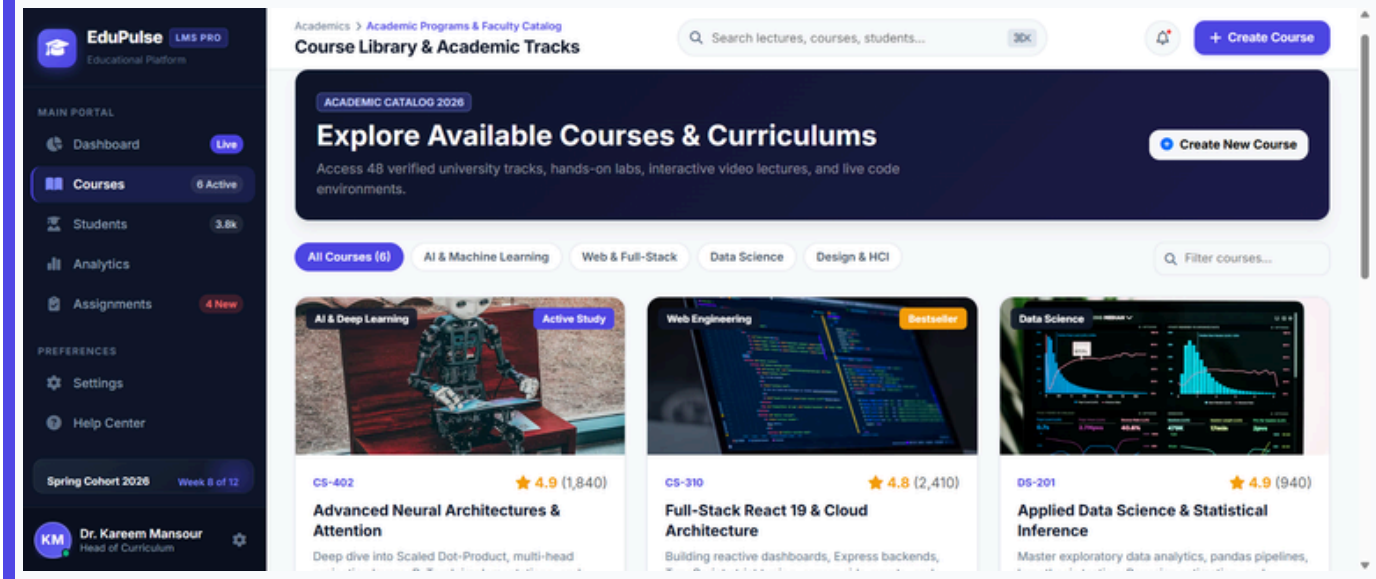
1. System Dashboard & Analytics Interface

• Overview: Serves as the primary operational hub, offering aggregated performance metrics and quick access to core learning resources.



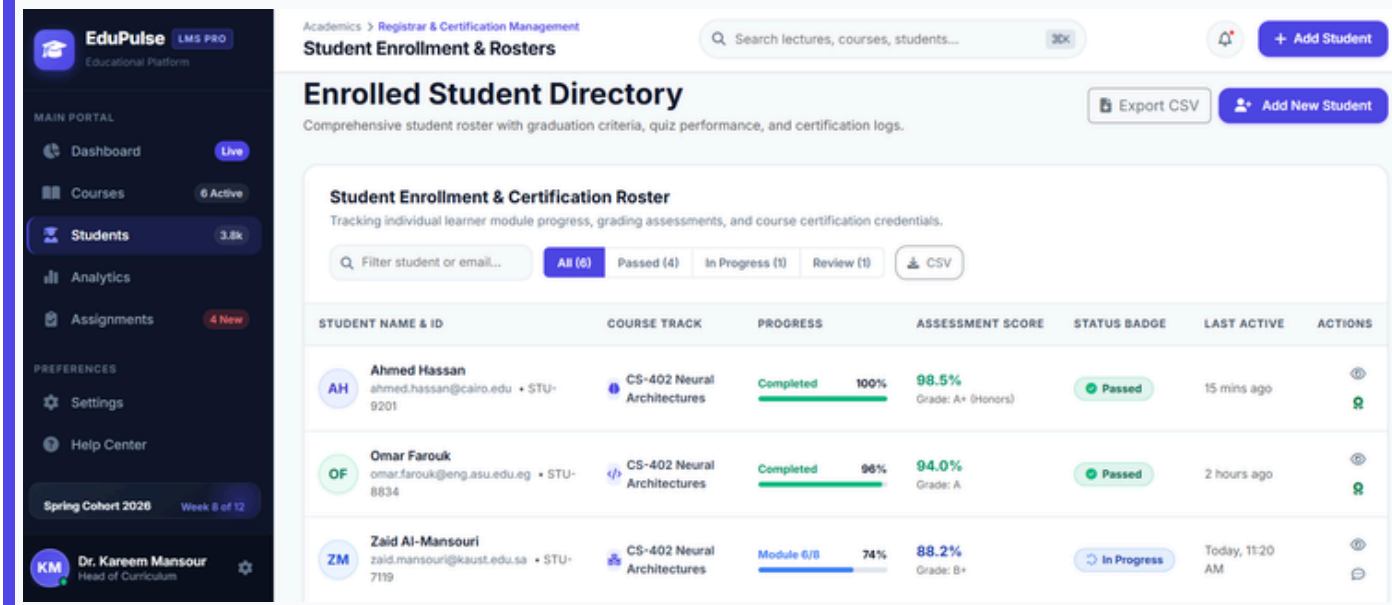
2. Course Library & Academic Catalog

• Overview: A grid-based catalog designed for searching, filtering, and discovering available academic tracks and active courses.



3. Student Enrollment & Directory (Data Matrix)

• Overview: An administrative management view optimized for monitoring learner progress, assessment evaluation, and certificate distribution.



4.2 UI/UX Guidelines & Prototyping Standards

Layout & Grid Framework: Built on Bootstrap 5.3 Flexbox Grid, ensuring fluid responsiveness across Desktop (>= 1200px), Tablet (>= 768px), and Mobile (>= 320px) viewports.

Color Palette: • Primary Brand: Deep Navy (#0F172A) utilized for persistent navigation sidebars, top application bars, and primary structural containers. • Accent Color: Vivid Indigo / Electric Blue (#4F46E5) reserved for primary call-to-action buttons, active navigation states, and progress indicators. • Success & State Indicators: Emerald Green (#10B981) dedicated to completed course progress, passed quiz badges, and positive performance metrics. • Surface & Backgrounds: Pure Neutral Light (#F8FAFC) base canvas paired with crisp white card bodies (#FFFFFF) and dark neutral text (#1E293B) for elevated contrast.

Typography: Inter and Segoe UI sans-serif typefaces prioritized for high legibility, strict contrast ratios meeting WCAG 2.1 AA specifications.

Component Standards: • Player Interface: Built on Video.js with floating controls, auto-buffering alerts, and synchronized side-drawer tabs for syllabus navigation and personal notes. • Asynchronous Feedback: Interactive elements communicate mutations via Toastr JS notifications and non-blocking spinner overlays. • Data Presentation: Administration matrices and student rosters leverage searchable, paginated DataTables with dynamic column sorting.